



**SIMATS ENGINEERING
CO3-ASSESSMENT TOOL -9**

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Course Code: CSA1511

Slot: B

Course Name: CLOUD COMPUTING AND BIG DATA ANALYTICS

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1. A company's cloud storage system requires 5 TB of data replication across three regions. What is the total storage required?

Answer: C) 15 TB

2. A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month?

Answer: A) ~4 minutes

3. A cloud-based load balancer distributes 60,000 requests per second among 10 servers. What is the average number of requests per server per second?

Answer: B) 6000

4. A company's cloud network has 1 Gbps bandwidth and needs to transfer 500 GB of data. How long will it take?

Answer: C) 1.1 hours

5. A cloud storage system experiences a 99.95% availability rate. How much downtime does it experience per year?

Answer: approximately 4.38 hours

6. A cloud provider offers pay-as-you-go pricing at \$0.05 per GB for storage. If a company stores 10 TB, what is the monthly cost?

Answer: B) \$500

7. A cloud network has 10 Gbps throughput but experiences 2% packet loss. What is the effective throughput?

Answer: A) 9.8 Gbps

8. A user uploads 250 MB of data per day to the cloud. How much data will be uploaded in one year (365 days)?

Answer: A) 91.25 GB

9. A cloud-based video streaming service requires 5 Mbps bandwidth per user. If 100 users are streaming simultaneously, what is the total required bandwidth?

Answer: A) 500 Mbps

10. A cloud provider guarantees 99.95% uptime per year. How much downtime does this allow in minutes per year?

Answer: A) 262 minutes

11. A virtualized data center has 200 physical servers. If each server runs 40 virtual machines, what is the total number of VMs in the data center?

Answer: B) 8,000

12. A physical server has 128 GB of RAM and hosts 10 virtual machines, each requiring 12 GB. How much memory is overcommitted?

Answer: A) 4 GB

13. A virtual machine requires 4 vCPUs. If a server has a total of 64 cores and an overcommitment ratio of 2:1, how many VMs can be hosted?

Answer: B) 32

14. A company migrates 50 on-premise servers to the cloud, reducing power consumption by 60%. If the original power usage was 100 kW, what is the new power consumption?

Answer: A) 40 kW

15. A cloud provider operates a virtualized data center with 100 physical servers, each with 64 vCPUs. If the overcommitment ratio is 4:1, how many total vCPUs can be allocated?

Answer: C) 25,600

16. Which cloud deployment model is used for organizations that require both private and public cloud resources?

Answer: A) Hybrid Cloud

17. Which cloud security strategy ensures that sensitive data remains private even if intercepted?

Answer: A) Encryption

18. If a hypervisor hosts 50 VMs and each VM uses 20 GB bandwidth per day, calculate total bandwidth consumption per week.

Answer: B) 7,000 GB

19. A cloud system runs at 70% average CPU utilization. If it has 32 cores at 2.5 GHz each, what is total utilized processing power?

Answer: A) 56 GHz

20. If the network latency in a virtualized system is 5 ms and processing time is 10 ms, calculate total round-trip delay for 100 requests.

Answer: A) 1.5 sec

21. If 100 VMs transfer logs of 200 MB/day each to the cloud, calculate the total data transferred in a 30-day month.

Answer: A) 600 GB

22. A cloud service's availability is 99.95%. How much downtime (in minutes) is expected in a month (30 days)?

Answer: A) 21.6 min

23. If a cloud server handles 200 requests/sec and each request uses 10 KB memory buffer, compute memory usage per minute.

Answer: A) 120 MB

24. A hypervisor hosts 120 VMs. If 75% of them require 2 vCPUs each, calculate total vCPU requirement.

Answer: A) 180 vCPUs

25. A company has 20 racks with 20 servers each. If each server uses 3 virtual NICs, how many vNICs exist in total?

Answer: A) 1,200

26. If a hypervisor can boot a VM in 2 minutes, how long will it take to boot 60 VMs in batches of 10 concurrently?

Answer: A) 12 min

27. A cloud data center uses 5 cooling units, each consuming 3.5 kW. Find total cooling energy per day.

Answer: A) 420 kWh

28. Each server is configured with 2 TB storage. A virtual SAN clusters 20 such servers. What is total usable capacity, assuming 20% is used for redundancy?

Answer: A) 32 TB

29. Which of the following is NOT a benefit of virtualization?

Answer: B) Increased energy consumption

30. Which feature in virtualization allows multiple OS environments to run simultaneously?

Answer: A) Hypervisor